



Department of Mathematics
IIT Hyderabad
MA1110: Calculus I
Credits: 1
Fall 2026

August 29, 2026

Time: 120 Minutes

Final Exam

Maximum Marks: 80

Name:

Roll No:

Instructions:

1. This question paper contains 20 **questions** in total, printed on 6 **pages**. Each question carries 4 **marks**.
2. There are four multiple-choice options in each question, followed by a “TRUE” and “FALSE” column. In the “TRUE” column, write those options (A/B/C/D) that are true (**in the box only**). Similarly, in the “FALSE” column, you have to write those options (A/B/C/D) that are not true (**in the box only**).
3. Any answers written outside the box or marked with a tick mark in the options will not be evaluated.
4. **Each correct marking will carry 1 mark. Each incorrect marking will carry a $-1/2$ mark. Unattended markings will carry 0 marks. Any option if placed in both columns will have the penalty of -1 mark.**
5. You are allowed to write the exam only after your **physical ID** card has been verified. Missing ID cards or digital ID cards will not be accepted.
6. Mobile phones, electronic gadgets, and digital watches are strictly prohibited during the examination. Any violation of these rules, including the use or possession of such devices, will result in an **F grade**, with no further discussion or consideration.

1. Let $f : [0, 2] \rightarrow \mathbb{R}$ be continuous and satisfy $f(0) = f(2)$. Then

- (A) There exists $c \in [0, 1]$ such that $f(c) = f(c + 1)$.
(B) There exists $c \in [0, 1]$ such that $f(c) - f(c + 1) = \max_{x \in [0, 1]} |f(x) - f(x + 1)|$.
(C) If $f(0) \neq f(1)$, then there exists $c \in (0, 1)$ such that $f(c) = f(c + 1)$.
(D) There exist $c_1, c_2 \in [0, 1]$ with $c_1 \neq c_2$ such that $f(c_1) = f(c_1 + 1) = f(c_2) = f(c_2 + 1)$.

TRUE:					FALSE:				
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2. Let $f(x)$ be a polynomial with real coefficients of degree 2025, and suppose that f has k distinct real roots. Consider the following statements. Select True options if the statement is necessarily true, or else choose False options.

- (A) f has at least one real root.
(B) f' has at least $k - 1$ distinct real roots.
(C) f' must have at least one real root.
(D) There exists $b \in \mathbb{R}$ such that $f(b) = f'(b)$.

TRUE:					FALSE:				
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3. Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable and suppose that $|g'(x)| \leq M$ for all $x \in \mathbb{R}$, where $M > 0$. For $\varepsilon > 0$, define

$$f_\varepsilon(x) := x + \varepsilon g(x).$$

Then

(A) There exists $\nu > 0$ such that f_ε is one-to-one for every $\varepsilon \in (0, 551\nu)$.

(B) If $\varepsilon > \frac{1}{M}$, then f_ε is not one-to-one.

(C) If $\varepsilon = \frac{1}{551M}$, then f_ε is one-to-one.

(D) If f_ε is one-to-one, then necessarily $\varepsilon \leq \frac{1}{M}$.

TRUE:					FALSE:				
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4. Let $f : [0, 1] \rightarrow [0, 1]$ be differentiable and suppose that

$$|f'(x)| \leq k < 1 \quad \text{for all } x \in [0, 1],$$

where k is a constant. Define

$$a_n = f\left(\frac{1}{n}\right), \quad n \in \mathbb{N}.$$

Then

(A) $0 \leq a_n^{2026} \leq 1$ for all $n \in \mathbb{N}$.

(B) The sequence $\{a_n\}_{n \geq 1}$ converges.

(C) The sequence $\{a_n\}_{n \geq 1}$ converges to the unique number $L \in [0, 1]$ satisfying $L = f(L)$.

(D) If $g : [0, 1] \rightarrow [0, 1]$ is another differentiable function with $|g'(x)| \leq k < 1$ for all $x \in [0, 1]$ and $g(0) = f(0)$, then the sequence $b_n = g\left(\frac{n^{2025} + \sqrt{n} + n \log(n)}{n^{2026}}\right)$ for $n \geq 2026$ converges to the same limit as $\{a_n\}_{n \geq 1}$.

TRUE:					FALSE:				
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5. For $a > 0$, consider

$$S_1(a) = \sum_{n=1}^{\infty} a^{\ln n}, \quad S_2(a) = \sum_{n=1}^{\infty} \frac{a^n \log n}{n^2}.$$

Then

(A) $S_1(a)$ and $S_2(a)$ converge if and only if $0 < a < e^{-1}$.

(B) $S_1(a)$ converges if and only if $0 < a \leq e^{-1}$.

(C) $S_2(a)$ converges if and only if $0 < a < 1$.

(D) $S_1(a)$ diverges whenever $S_2(a)$ converges.

TRUE:					FALSE:				
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6. Let $\{a_n\}_{n \geq 1}$ be a sequence of positive real numbers satisfying $a_1 \geq 1$ and $a_n \geq a_{n+1}$ for all $n \geq 1$. Then

(A) The series $\sum_{n=1}^{\infty} \frac{1}{a_n}$ diverges.

(B) The sequence $\{a_n\}_{n \geq 1}$ is bounded.

(C) The series $\sum_{n=1}^{\infty} \frac{1}{a_n^2}$ converges.

(D) The series $\sum_{n=1}^{\infty} \frac{1}{a_n}$ converges.

TRUE:					FALSE:				
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7. If $s_1 > 0$ and $s_{n+1} = \frac{1}{2} \left(s_n + \frac{4}{s_n} \right)$, $n \geq 1$, then

(A) The sequence $\{s_n\}$ is monotone decreasing for $n \geq 2$.

(B) $\{s_n\}_{n \geq 1}$ is bounded below.

(C) $\lim_{n \rightarrow \infty} s_n = 2$.

(D) The sequence $\{s_n\}_{n \geq 1}$ diverges.

TRUE:					FALSE:				
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8. Let $\{u_n\}_{n \geq 1}$ and $\{v_n\}_{n \geq 1}$ be sequences of real numbers. Then

(A) If $\{u_n^2\}_{n \geq 1}$ converges to $L > 0$, then $\{u_n\}_{n \geq 1}$ converges.

(B) If $u_n \rightarrow 0$ and $\{v_n\}_{n \geq 1}$ is bounded, then $\{u_n v_n\}_{n \geq 1}$ converges to 0.

(C) If $\{u_n\}_{n \geq 1}$ is a Cauchy sequence and $|u_n - v_n| \rightarrow 0$, then $\{v_n\}_{n \geq 1}$ is also a Cauchy sequence.

(D) There exist divergent sequences $\{u_n\}_{n \geq 1}$ and $\{v_n\}_{n \geq 1}$ such that

$$u_{2n} = v_{2n} \quad \text{for all } n \geq 1$$

and

$$|u_{2n+1} - v_{2n+1}| \rightarrow 0.$$

TRUE:					FALSE:				
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9. Let $\{u_n\}_{n \geq 1}$ be a sequence of positive real numbers such that $u_n \rightarrow l$, where $l > 0$. Then

(A) The series $\sum_{n=1}^{\infty} \frac{u_n^2}{n^3}$ always converges.

(B) The series $\sum_{n=1}^{\infty} \frac{u_n}{n}$ converges for at least one such sequence.

(C) If $l < 1$, then the series $\sum_{n=1}^{\infty} u_n^{n/2}$ converges.

(D) If $l < 1$, then the series $\sum_{n=1}^{\infty} \frac{1}{u_n^n}$ converges.

TRUE:					FALSE:				
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10. For each of the following functions, select True options if the function is continuous on its respective domain, or else choose False options.

(A) $f : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ defined by $f(x) = e^{1/x}$.

(B) $f : [0, 4] \rightarrow \mathbb{R}$ defined as $f(x) = \begin{cases} \frac{x^2-4}{x-2} & \text{if } x \neq 2 \\ 5 & \text{if } x = 2 \end{cases}$.

(C) $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as $f = h \circ g$ where $g(x) = \text{sgn}(x)$ and $h(x) = x(1 - x^2)$ on $\mathbb{R}$.

(D) $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as $f = g \circ h$ where $g(x) = \text{sgn}(x)$ and $h(x) = x(1 - x^2)$ on $\mathbb{R}$.

TRUE:					FALSE:				
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11. For each of the following conditions, select True options if the condition ensures the convergence of the real sequence $\{a_n\}_{n \geq 1}$, or else choose False options.

(A) $|a_n - a_{n+1}| \rightarrow 0$ as $n \rightarrow \infty$.

(B) $\sum_{n=1}^{\infty} |a_n - a_{n+1}|$ is convergent.

(C) $\sum_{n=1}^{\infty} na_n$ is convergent.

(D) The sequence $\{a_{3n}\}_{n \geq 1}$, $\{a_{3n+1}\}_{n \geq 1}$ & $\{a_{3n+2}\}_{n \geq 1}$ are convergent.

TRUE:					FALSE:				
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12. Let

$$f(x) = \begin{cases} 3x^2, & x \in \mathbb{Q} \\ 0, & x \in \mathbb{Q}^c \end{cases}$$

Then f is

(A) Continuous but not differentiable at $x = 0$.

(B) Continuous but not differentiable at $x = 1$.

(C) Differentiable at $x = 1$.

(D) Differentiable at $x = 0$.

TRUE:					FALSE:				
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13. If $\sum_{n=1}^{\infty} a_n$ & $\sum_{n=1}^{\infty} b_n$ be two convergent series, then

(A) The series $\sum_{n=1}^{\infty} a_n^2$ is also convergent.

(B) The series $\sum_{n=1}^{\infty} a_n b_n$ is also convergent.

- (C) The series $\sum_{n=1}^{\infty} \frac{a_n}{1+a_n}$ is divergent, where in addition assume $a_n \neq -1$ for all $n \geq 1$.
- (D) The series $\sum_{n=1}^{\infty} (a_n + b_n)^2$ is also convergent.

TRUE:					FALSE:				
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14. Let $\{x_n\}_{n \geq 1}$ be a sequence of real numbers such that $\lim_{n \rightarrow \infty} (x_{n+1} - x_n) = c$, where $c > 0$. Then the sequence $\left\{\frac{x_n}{n}\right\}_{n \geq 1}$
- (A) is not bounded.
- (B) is bounded but does not converge.
- (C) converges to c .
- (D) converges to 0.

TRUE:					FALSE:				
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15. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x+y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$. For each of the following conditions, select True options if the condition is sufficient to guarantee that f is continuous everywhere, and select False options otherwise.
- (A) f is continuous at $x = 0$.
- (B) $f(0) = 0$.
- (C) f is differentiable at $x = 551$.
- (D) f is continuous at all irrational points in $\mathbb{R}$.

TRUE:					FALSE:				
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16. Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by:

$$f(x) = \begin{cases} x & \text{if } x \in \mathbb{Q} \\ 2026 - x & \text{if } x \in \mathbb{Q}^c \end{cases}$$

Then

- (A) f is discontinuous everywhere.
- (B) f is continuous only at $x = 1013$.
- (C) f is continuous on $\mathbb{Q}$ and discontinuous on $\mathbb{R} \setminus \mathbb{Q}$.
- (D) f is continuous at $x = 2026$ and $x = 0$.

TRUE:					FALSE:				
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17. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable everywhere and satisfies $f'(x) > f(x)$ for all $x \in \mathbb{R}$. If $f(0) = 1$, then
- (A) $f(x) < e^x$ for all $x > 0$.
- (B) $f(x) > e^x$ for all $x > 0$.

- (C) $f(x) > e^x$ for all $x < 0$.
 (D) $f(x) = e^x$ has at least one real solution for $x > 0$.

TRUE:					FALSE:				
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18. Let $\{a_n\}_{n \geq 1}$ be a sequence of real numbers. Select True options if the following condition ensures that $\{a_n\}_{n \geq 1}$ is a Cauchy sequence, or else choose False options.

- (A) $|a_{n+1} - a_n| \rightarrow 0$ as $n \rightarrow \infty$
 (B) $|a_{n+1} - a_n| < \frac{1}{n}$ for all $n \geq 1$
 (C) $|a_{n+1} - a_n| < \frac{1}{2^n}$ for all $n \geq 1$
 (D) There exists $N \in \mathbb{N}$ such that $|a_{n+1} - a_n| < \frac{1}{n^2}$ for all $n \geq N$.

TRUE:					FALSE:				
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19. Let $\sum_{n=1}^{\infty} a_n$ be a conditionally convergent series of real numbers. Define $p_n = \frac{a_n + |a_n|}{2}$ (positive terms) and $q_n = \frac{a_n - |a_n|}{2}$ (negative terms). Then

- (A) Both $\sum_{n=1}^{\infty} p_n$ and $\sum_{n=1}^{\infty} q_n$ converge.
 (B) $\sum_{n=1}^{\infty} p_n$ converges, but $\sum_{n=1}^{\infty} q_n$ diverges.
 (C) Both $\sum_{n=1}^{\infty} p_n$ and $\sum_{n=1}^{\infty} q_n$ diverge.
 (D) $\sum_{n=1}^{\infty} p_n$ diverges, but $\sum_{n=1}^{\infty} q_n$ converges.

TRUE:					FALSE:				
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20. Let $f : [0, 5] \rightarrow \mathbb{R}$ be continuous on $[0, 5]$ and differentiable on $(0, 5)$. Suppose that $f(0) = -3$ and $f'(x) \leq 4$ for all $x \in (0, 5)$. Also, consider the real roots of $e^x \cos x = 0$. Select True options if the following statement is necessarily true, or else choose False options.

- (A) $f(5) \leq 17$, and equality can occur.
 (B) $f(5) < 17$ for every function satisfying the given conditions.
 (C) Between any two consecutive real roots of $e^x \cos x = 0$, there is exactly one real root of $e^x \sin x = 0$.
 (D) Between any two consecutive real roots of $e^x \sin x = 0$, there are exactly two real roots of $e^x \cos x = 0$.

TRUE:					FALSE:				
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ID card & Signature of Student

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Invigilator Short Sign

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TA Signature Checking

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